Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/14/2025
PI Name:	Settles, Andrew Mark Ph.D.		
Project Title:	Understanding Genome-Wide Mutation Load in Spaceflight Culture of Cyanobacteria		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	andrew.m.settles@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	352-283-2767
Organization Name:	NASA Ames Research Center		
PI Address 1:	AST Life Support Studies		
PI Address 2:	NASA Ames Research Center		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	10/01/2021	**End Date:**	09/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2026 per F. Hernandez/ARC (Ed.,2/21/26). NOTE: End date changed to 09/30/2025 per F. Hernandez/ARC (Ed., 9/19/24). NOTE: End date changed to 09/30/2024 per F. Hernandez/ARC (Ed., 10/10/23). NOTE: Project dates for this investigation have changed; the period of performance (POP) is now 10/1/2021-09/30/2023 per F. Hernandez/ARC. Original POP was 09/1/2021 - 08/31/2024 (Ed. 1/19/22).		
Key Personnel Changes/Previous PI:	Nothing to report		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Increased radiation and the microgravity environment of spaceflight is expected to increase genetic mutation of all organisms. Biological Life Support systems incorporate organisms that can fix carbon dioxide and nitrogenous wastes into sugars, proteins, and other carbohydrates to release oxygen and provide feedstocks for food, fuel, or manufacturing. Plants and microbes can be used for biological support, and it is essential to understand how spaceflight impacts mutation rates in these organisms to enable appropriate countermeasures to ensure high life support productivity during long duration and deep space missions. This project will focus on the cyanobacterium Arthrospira platensis. Commonly known as Spirulina, this is an edible microalgae that is a candidate for biological life support. The project will grow A. platensis in serial cultures to allow the organism to evolve in spaceflight on the International Space Station. Whole genome sequencing of culture samples will determine the mutation load of the cyanobacteria at multiple time points throughout the long-term growth experiment. Multi-omics analysis including RNA sequencing, proteomics, metabolomics, and analytical chemistry of nutritional composition will give insights on the biological responses of the cells to spaceflight. The project directly addresses research topic 1 (Microbiology Studies in Support of Space Exploration) by studying a microorganism that is a candidate to support long-duration space exploration. The experiments proposed will reveal whether long-term growth of A. platensis will be simple to implement or if there is risk of undesirable cellular responses that require mitigation. If spaceflight is a significant stress for this species, the project will identify specific genes and pathways that would need to be targeted for genetic improvement. The project is expected to further our knowledge of microorganism adaptation to long-term growth in spaceflight and contribute towards the development of bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project primarily investigates DNA damage risks associated with low Earth orbit. The focus is on an algae crop that is grown as a protein source in human and animal diets. The fundamental knowledge developed in the project will contribute to a better understanding of DNA damage mechanisms and may reveal new targets for protecting humans or agriculture from environmental stresses.
Task Progress:	We completed flight definition experiments to refine our science requirements and concept of operations for the flight experiment. Our primary focus this year was on testing cryopreservation scenarios to deliver viable frozen cells to the International Space Station (ISS) and to return viable cells from each growth cycle planned for the on-orbit operations. Our testing has demonstrated a variety of biocompatible vessels can be used to cryopreserve Arthrospira platensis (A. platensis) strain NIES-39. These results enable a more flexible accommodation of Space Algae-2 on a variety of flight and operations scenarios. Demonstration of multiple variations in cryopreservation methods allow flexible experiment design for launch and on-orbit cryopreservation. Compared to many biological experiments that are metabolically active during launch and transit to the ISS, cryopreserved cells allow for flexible timing of experiment initiation. The long-term viability of cryopreserved A. platensis reduces risk for bioregenerative life support systems. In the event of contamination of a spirulina production system in space, a cryopreserved stock could be used to restart production after a decontamination procedure. Without cryopreserved stocks, a separate spirulina stock culture would need to be continuously cultivated, in addition to a biological production system.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Peer-reviewed Journals	Fisher KE, Settles AM. "Efficient long-term cryopreservation of Arthrospira platensis NIES-39 and adaptation for the Space Algae-2 spaceflight experiment." npjMicrogravity. 2026 Apr 22. https://doi.org/10.1038/s41526-026-00598-w ; PMID: 42020394 , Jan-2026